Hospital Database Design Document

Version 1.0, Revision 2

Version History

Version	Description
1.0, rev 0	First released draft
1.0, rev 1	Added relational schema
1.0, rev 2	• Added database dictionary – DOB, admitDate, dischargeDate: nvarchar to smalldatetime • Added database diagram

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Narrative Summary

The hospital administrator wants to create a database to track nurse assignments to their wards and nurse interactions with their patients, patient admissions by their doctors and treatments administered by doctors to their patients, bed assignments for each patient and items charged to patients during their stay. Administrator wants to record each nurse's name and address, phone and alternate phone, email and the medical specialties he or she is certified. Some nurses supervise one or more other nurses. No nurse is supervised by more than one nurse, and some nurses are unsupervised.

Each ward at the hospital has a designated number, descriptive name, physical location and phone number. Each ward has at least one nurse assigned to it. A nurse is assigned to at least one ward and rotates assignments among other wards. The assignment is tracked by the specific date and the hours worked in the assigned ward by each nurse on that date.

In addition to nurse assignments, each ward also has a charge nurse. The charge nurse is the custodian of the medical records for the ward. Not all nurses act in this capacity, but those that do are in charge of only one ward, and a ward only has one charge nurse.

A ward consists of hospital beds. The beds are inventoried to a specific ward. Information on beds including their size (small, large, extra-large) and their type (elevated electrically or manually). Most of the beds are large and manual (this is the default setting).

When a patient is admitted to the hospital they are assigned to a specific bed. Not all beds are available for use all the time, and a bed may not be assigned to more than one patient.

Information on patients is recorded: name, gender, dob, address, phone, alternate phone, email. The date the patient is admitted to the hospital, the admitting doctor, the date the patient is discharged, and discharging doctor are also tracked.

Some doctors admit patients while others do not. Doctor information tracked: name, address, phone, alternate phone, email and their medical specialties.

The hospital tracks the treatments administered to patients and the treating doctor. Treatments are tracked by name, description, and charge. The hospital also tracks the

date and time of each treatment administered and the results. Some doctors treat patients while others do not.

A given patient may receive no treatments or may receive many, and some patients may receive their treatments from more than one doctor. Some treatments have yet to be used while others have been used often.

In addition to treatments, patients incur other charges for items used during their stay. The hospital tracks these charges as “items” and stores information on what items have been charged to which patients, based on date and quantity. Information that is to be stored for each item includes the item name and charge. All patients incur at least one charge for consumable items used during their stay. Some items are used often while items may be new or unusual in nature and might rarely or never be charged to any patients.

Lastly, the hospital tracks nurse patient care. Each nurse patient care interaction is an event. There are several types of events: wellness check, medication, food service, assistance, treatment admin, and “other.” Given the number of shifts and ward rotations, a patient will typically be seen by more than one nurse during their stay, and a nurse most likely will interact with the same patient over several events during a single shift.

Actors & Roles

- Nurse: Interacts with patient, conducting events such as wellness checks. Charge nurses oversee their individual ward's medical records.
- Ward: A nurse is assigned to a ward with assignments rotating along the collection of wards. Each ward is led by a single charge nurse.
- Patient: Admitted into a hospital by a doctor and assigned a single bed.
- Doctor: Admits patients and administers treatment.
- Treatment:
- Bed: Many beds are inventoried to a specific ward. Also assigned to one patient per bed.
- Item: Charges incurred by patients during their stay: i.e. usage of a consumable item.

List of Entities

- Nurse
- Ward
- Patient
- Doctor
- Treatment
- Bed
- Item

List of Entities (with nested attributes)

- Nurse
    - Name
    - Address
    - Phone
    - Alternate phone
    - Email
    - Medical specialty
- Ward
    - Number
    - Descriptive name
    - Physical location
    - Phone number
    - Nurse(s)
    - Charge nurse
    - Bed
- Patient
    - Name
    - Gender
    - DOB
    - Address
    - Phone
    - Alternate phone
    - Email
    - Admittance date
    - Admitting doctor
    - Discharge date
    - Discharging doctor
- Doctor
    - Name

    - Address
    - Phone
    - Alternate phone
    - Email
    - Medical specialty(ies)

- Treatment
    - Name
    - Description
    - Charge
    - Date administered
    - Effects
- Bed
    - Size
    - Type
- Item
    - Name
    - Charge

Business Rules

Nurse: Nurses supervise many patients. One nurse oversees many nurses and their ward's medical records.

Ward: Wards are assigned many beds and staffed by many nurses.

Patient: Patients receive one bed and are cared for by many nurses. Patients receive one, more than one, or no treatments and use one, more than one, or no items.

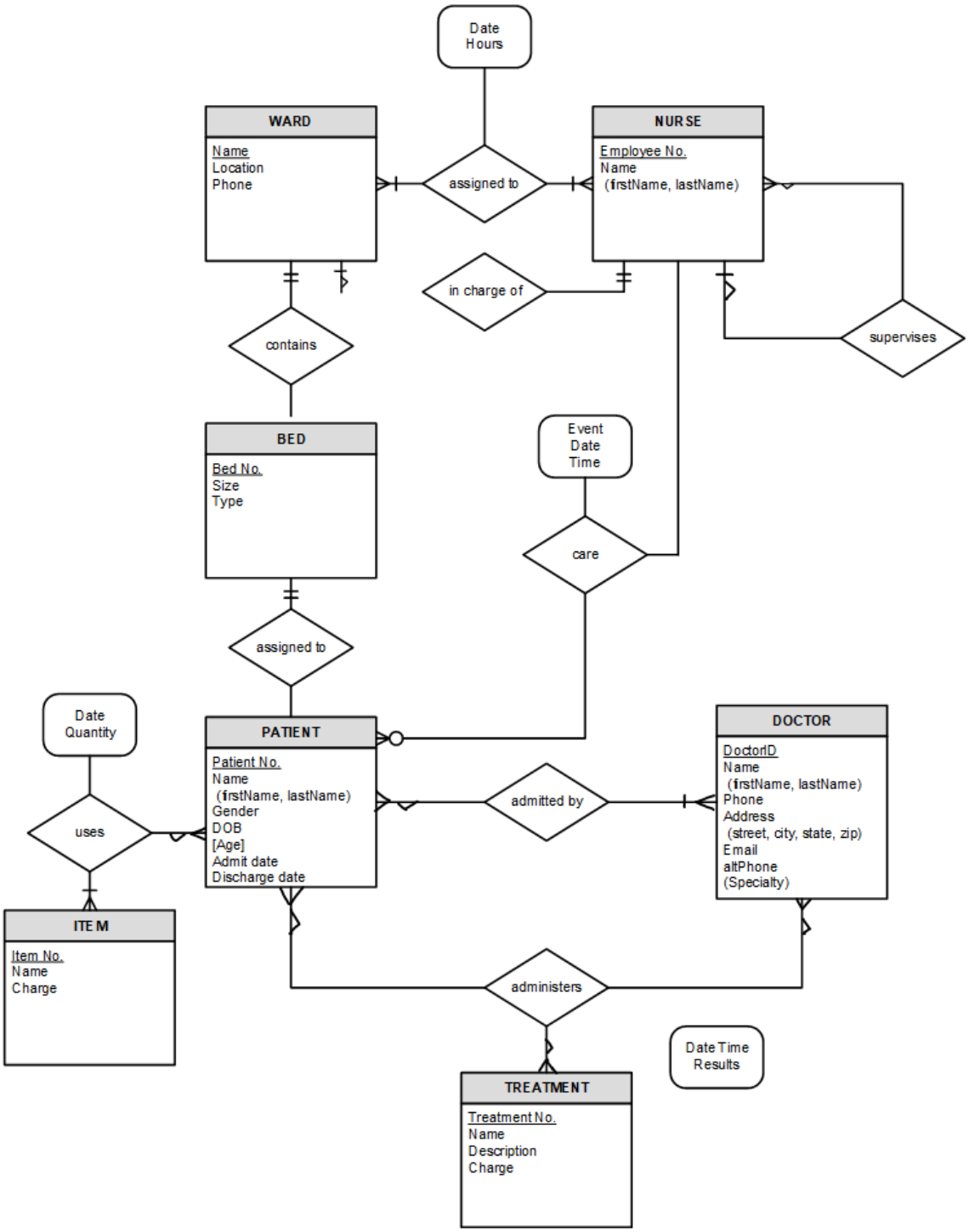
Doctor: Doctors admit one, none, or more than one patient.

Treatment: Treatments are administered to patients by doctors.

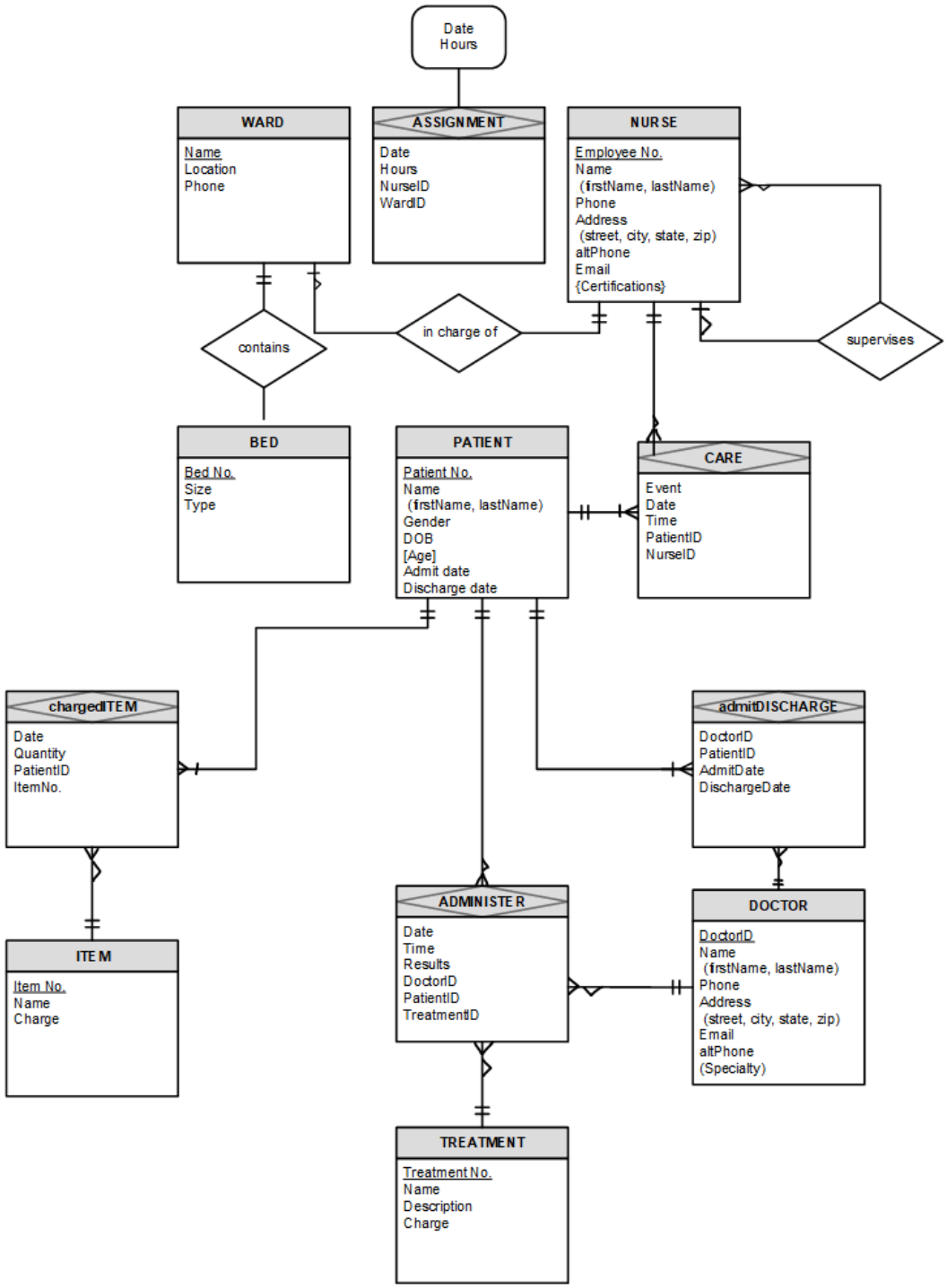
Bed: Beds are assigned to one and only one ward.

Item: Items are charged to patients once or more than once.

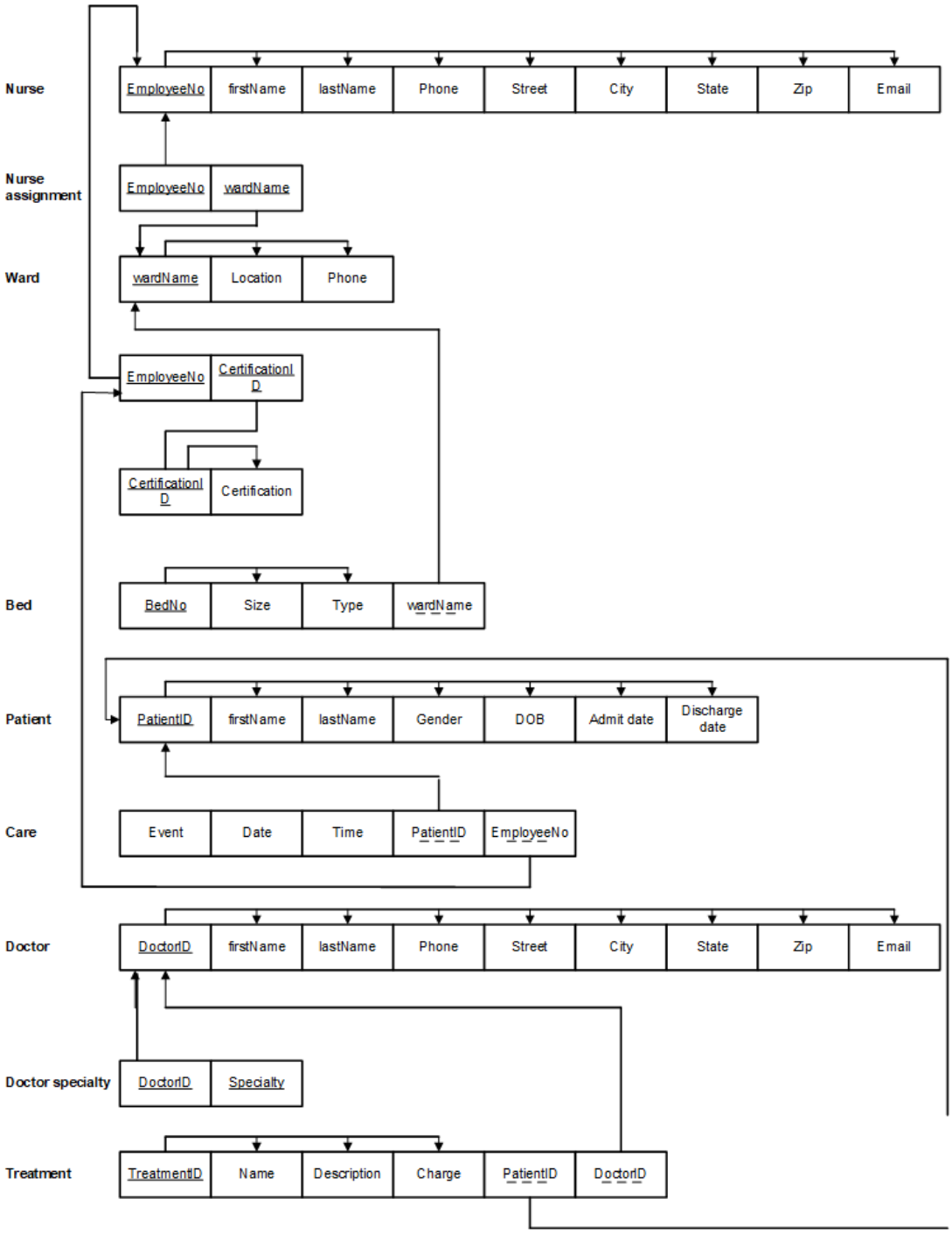
ERD


Date
Hours
WARD
Name
Location
Phone
assigned to
NURSE
Employee No.
Name
(firstName, lastName)
in charge of
contains
supervises
BED
Bed No.
Size
Type
Event
Date
Time
care
assigned to
Date
Quantity
PATIENT
Patient No.
Name
(firstName, lastName)
Gender
DOB
[Age]
Admit date
Discharge date
admitted by
DOCTOR
DoctorID
Name
(firstName, lastName)
Phone
Address
(street, city, state, zip)
Email
altPhone
(Specialty)
uses
ITEM
Item No.
Name
Charge
administers
Date Time
Results
TREATMENT
Treatment No.
Name
Description
Charge

EERD


Date
Hours
WARD
Name
Location
Phone
ASSIGNMENT
Date
Hours
NurseID
WardID
NURSE
Employee No.
Name
(firstName, lastName)
Phone
Address
(street, city, state, zip)
altPhone
Email
{Certifications}
contains
in charge of
supervises
BED
Bed No.
Size
Type
PATIENT
Patient No.
Name
(firstName, lastName)
Gender
DOB
[Age]
Admit date
Discharge date
CARE
Event
Date
Time
PatientID
NurseID
chargedITEM
Date
Quantity
PatientID
ItemNo.
admitDISCHARGE
DoctorID
PatientID
AdmitDate
DischargeDate
ITEM
Item No.
Name
Charge
ADMINISTER
Date
Time
Results
DoctorID
PatientID
TreatmentID
DOCTOR
DoctorID
Name
(firstName, lastName)
Phone
Address
(street, city, state, zip)
Email
altPhone
(Specialty)
TREATMENT
Treatment No.
Name
Description
Charge

Relational Schema



Data Dictionary

Table: **Bed**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
BedNo	**PK**; Unique sequential bed number	int		Y	Y					Y
Size	Size of the bed	nvarchar	15			Large	Small, medium, large, extralarge			
Type	Type of bed	nvarchar	20			Manual	Electric, manual			

Table: **Care**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index

Event	Interactions between nurse and patients	nvarchar	15							Y
Date	Date care event took place	small datetime								
Time	Time of care event	nvarchar								
PatientID	**FK** to patient table	int							Y	
EmployeeNo	**FK** to nurse table	int								

Table: **Doctor**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
DoctorID	**PK**; Unique sequential doctor ID number	int		Y	Y					Y
firstName	First name of the doctor	nvarchar	15							
lastName	Last name of the doctor	nvarchar	20							
Phone	Phone number of the employee	nvarchar	20						Y	
Street	Street of the employee	nvarchar	20							
City	City of the employee	nvarchar	15							
State	State of the employee	char	2					LIKE ‘[A-Z] [A-Z]’		
Zip	Zip code of the employee	nvarchar	10					([Zip] LIKE '[0-9][0-9][0-9][0-9][0-9]')		
Email	Email of the doctor	nvarchar	70							
Specialty	Doctor’s specialty(ies)	nvarchar								

Table: **Nurse**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
EmployeeNo	**PK**; Unique sequential nurse number	int		Y	Y					Y
firstName	First name of the nurse	nvarchar	15							
lastName	Last name of the nurse	nvarchar	20							
Phone	Phone number of the nurse	nvarchar	20						Y	
Street	Street of the nurse	nvarchar	20							
City	City of the nurse	nvarchar	15							
State	State of the nurse	char	2					LIKE ‘[A-Z] [A-Z]’		
Zip	Zip code of the nurse	nvarchar	10					([Zip] LIKE '[0-9][0-9][0-9][0-9][0-9]')		
Email	Email of the nurse	nvarchar	70							

Table: **Nurse assignment**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
EmployeeNo	**FK** to nurse table	int								
wardName	**FK** to ward name	nvarchar	20							

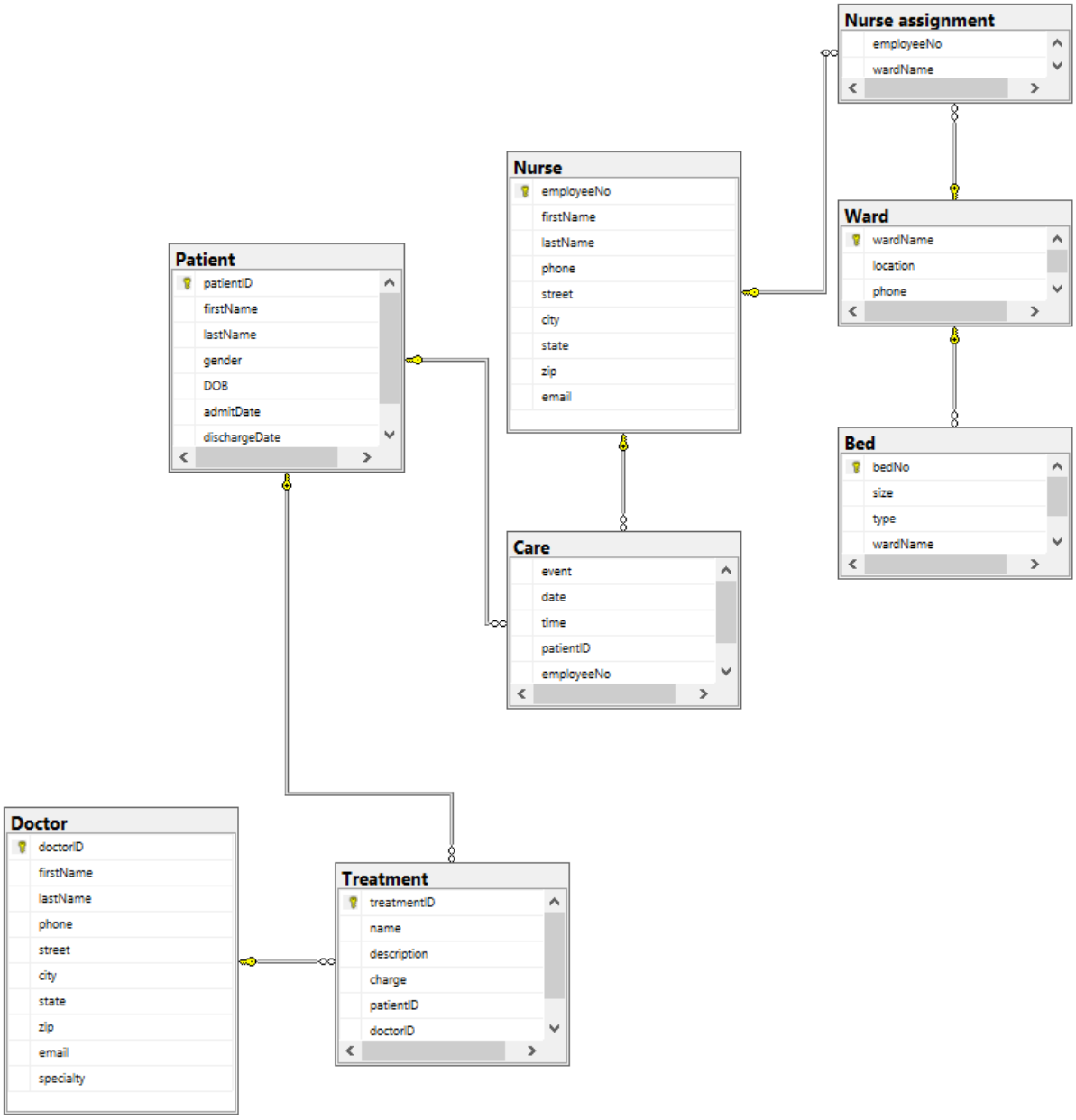
Table: **Patient**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
PatientID	**PK**; Unique sequential patient ID number	int		Y	Y					Y
firstName	First name of the patient	nvarchar	15							
lastName	Last name of the patient	nvarchar	20							
Gender	Gender of patient	char	1				M(ale), F(emale), O(ther)			
DOB	Patient date of birth	smalldatetime								
Admit date	Date of patient admittance	smalldatetime								
Discharge date	Date of patient discharge	smalldatetime							Y	

Table: **Treatment**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
treatmentID	**PK**; Unique sequential treatment ID number	int		Y	Y					Y
Name	Treatment name	nvarchar	15							
Description	Treatment description	nvarchar	20							
Charge	Charge incurred by patient	money								
PatientID	**FK** to patient table	int								
DoctorID	**FK** to doctor table	int								

Table: **Ward**

Column Name	Description	Data Type	Size	Identity	Unique	Default	Rule	Check	Allow Nulls	Index
wardName	**PK**; Unique ward name	nvarchar	20	Y	Y					Y
Location	Location of ward	nvarchar	15							
Phone	Ward phone number	nvarchar	20						Y	

Database Diagram